EXECUTIVE SUMMARY

This project focused on predicting hotel booking cancellations using the Hotel Bookings dataset, with the primary objective of demonstrating the importance of feature engineering over complex modeling techniques. Across all experiments, it was evident that improvements in data preprocessing and feature design had a significantly greater impact on model performance than simply changing algorithms.

The most important factor influencing model performance was feature engineering. While the baseline model with minimal preprocessing achieved moderate performance, substantial improvements were observed after applying numeric preprocessing, feature extraction, and feature construction. Techniques such as scaling, handling missing values, and binning helped stabilize the model, but the largest gains came from creating meaningful features that captured customer behavior and booking patterns. This highlights that understanding the data and transforming it effectively is more critical than using sophisticated models.

The key change that improved the performance ceiling was the introduction of constructed and aggregated features. Features such as price_per_person, total_nights, and guests_nights provided deeper insights into customer commitment and spending behavior. Similarly, interaction features like adr × lead_time captured non-linear relationships that were not visible in the raw data. Aggregated features such as average ADR by country and hotel type helped incorporate group-level behavior, further enhancing predictive power. Feature selection also played an important role by removing redundant and less informative features, leading to better generalization and reduced overfitting.

From a business perspective, several features emerged as highly actionable. The most impactful feature was lead_time, which indicates how early a booking was made; customers with longer lead times are more likely to cancel, allowing the business to target them with reminders or flexible policies. ADR (Average Daily Rate) is another critical feature, enabling pricing strategies such as dynamic discounts to reduce cancellations. Features like total_nights and guests_nights reflect booking commitment and can help identify high-value customers. Additionally, special_requests_rate indicates customer engagement, which can be used to prioritize personalized services.

In conclusion, this project demonstrates that feature engineering is the key driver of model performance, improving not only accuracy but also interpretability and business relevance. By focusing on meaningful feature creation and selection, the model becomes more reliable and provides actionable insights that can directly support business decisions such as targeted marketing, pricing optimization, and customer retention strategies.